

0.1.4 Inverse functions

A function is a rule for associating an output to an input.

Can we reverse this? Given an output y , can we find an x such that $f(x) = y$?

Example: Let $f(x) = x^3 + 4$ and $y = 5$.

What input x gives output 5?

$$\rightarrow x^3 + 4 = 5 \Leftrightarrow x^3 = 1 \Leftrightarrow x = \sqrt[3]{1} = 1$$

What about an output y ? Find a formula:

$$x^3 + 4 = y \Leftrightarrow x^3 = y - 4 \Leftrightarrow x = \sqrt[3]{y - 4}$$

Observation: This is a function! (of y)

Definition: Given a function f with domain D and range R , the inverse of f (if it exists) is the function f^{-1} with domain R and range D such that $f(x) = y$ if and only if $f^{-1}(y) = x$.

Equivalently, $f^{-1}(f(x)) = x$ and $f(f^{-1}(y)) = y$.

Example: Find the inverse of $f(x) = 7x + 1$.

$$\rightarrow 7x + 1 = y \Leftrightarrow 7x = y - 1 \Leftrightarrow x = \frac{y - 1}{7}$$

$f^{-1}(x) = \frac{x - 1}{7}$ (variable name doesn't matter, this is the same function as $f^{-1}(s) = \frac{s - 1}{7}$)

$$f^{-1}(f(x)) = \frac{f(x) - 1}{7} = \frac{7x + 1 - 1}{7} = \frac{7x}{7} = x$$

$$f(f^{-1}(x)) = 7f^{-1}(x) + 1 = 7\left(\frac{x - 1}{7}\right) + 1 = x - 1 + 1 = x$$

Example: Find the inverse of $f(x) = \frac{4x+1}{x}$.

$$\rightarrow \frac{4x+1}{x} = y \iff 4x+1 = xy \iff 4x - yx + 1 = 0$$

$$\iff 4x - yx = -1 \iff (4-y)x = -1 \iff x = \frac{-1}{4-y} = \frac{1}{y-4}$$

Then $f^{-1}(x) = \frac{1}{x-4}$.

Exercise: show that $f^{-1}(f(x)) = x$

and $f(f^{-1}(x)) = x$.

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$$\frac{1}{f(x)-4} = \frac{1}{\frac{4x+1}{x}-4} = \dots = x.$$